



TRIOS CYBER

LABORATORY PRACTICE 14

Local Web Log Analysis and Request Pattern Detection

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1. Apache Installation on Ubuntu

Apache2 was installed on the Ubuntu laboratory virtual machine with the APT package manager. The installation command downloaded and configured the Apache2 server and its required packages, preparing the endpoint to provide a local HTTP service for the web-log analysis activity.

```
matthew@matthew-VirtualBox:~$ sudo apt install apache2 -y
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias
Leyendo la información de estado... Hecho
Se instalarán los siguientes paquetes adicionales:
apache2-bin apache2-data apache2-utils libapr1 libaprutil1
libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.2-0
Paquetes sugeridos:
apache2-doc apache2-suexec-pristine | apache2-suexec-custom
Se instalarán los siguientes paquetes NUEVOS:
apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1
libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.2-0
0 actualizados, 0 nuevos se instalarán, 0 para eliminar y 501 no actualizados.
1 no instalados del todo o eliminados.
Se necesita descargar 1.828 kB de archivos.
Se utilizarán 7.976 kB de espacio de disco adicional después de esta operación.
Des1 http://ec.archive.ubuntu.com/ubuntu focal-updates/main amd64 libapr1 amd64 1.6.5-1ubuntu1.1 [91,5 kB]
Des2 http://ec.archive.ubuntu.com/ubuntu focal-updates/main amd64 libaprutil1 amd64 1.6.1-4ubuntu2.2 [85,1 kB]
Des3 http://ec.archive.ubuntu.com/ubuntu focal-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.1-4ubuntu2.2 [105,5 kB]
Des4 http://ec.archive.ubuntu.com/ubuntu focal-updates/main amd64 libaprutil1-ldap amd64 1.6.1-4ubuntu2.2 [8.752 B]
Des5 http://ec.archive.ubuntu.com/ubuntu focal/main amd64 liblua5.2-0 amd64 5.2.4-1.1build3 [100 kB]
Des6 http://ec.archive.ubuntu.com/ubuntu focal-updates/main amd64 apache2-bin amd64 2.4.41-4ubuntu3.23 [1.188 kB]
Des7 http://ec.archive.ubuntu.com/ubuntu focal-updates/main amd64 apache2-data all 2.4.41-4ubuntu3.23 [159 kB]
Des8 http://ec.archive.ubuntu.com/ubuntu focal-updates/main amd64 apache2-utils amd64 2.4.41-4ubuntu3.23 [24,4 kB]
Des9 http://ec.archive.ubuntu.com/ubuntu focal-updates/main amd64 apache2 amd64 2.4.41-4ubuntu3.23 [95,6 kB]
Descargados 1.828 kB en 2s (1.038 kB/s)
Seleccionando el paquete libapr1:amd64 previamente no seleccionado.
(Leyendo la base de datos ... 224981 ficheros o directorios instalados actualment
e.)
Preparando para desempaquetar .../0-libapr1-1.6.5-1ubuntu1.1-amd64.deb ...
Desempaquetando libapr1:amd64 (1.6.5-1ubuntu1.1) ...
Seleccionando el paquete libaprutil1:amd64 previamente no seleccionado.
Preparando para desempaquetar .../1-libaprutil1-1.6.1-4ubuntu2.2-amd64.deb ...
Desempaquetando libaprutil1:amd64 (1.6.1-4ubuntu2.2) ...
Seleccionando el paquete libaprutil1-dbd-sqlite3:amd64 previamente no seleccionado.
Preparando para desempaquetar .../2-libaprutil1-dbd-sqlite3-1.6.1-4ubuntu2.2-amd64.deb ...
Desempaquetando libaprutil1-dbd-sqlite3:amd64 (1.6.1-4ubuntu2.2) ...
Seleccionando el paquete libaprutil1-ldap:amd64 previamente no seleccionado.
```

```
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /lib/s
ystemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /lib/s
ystemd/system/apache-htcacheclean.service.
Procesando disparadores para ufw (0.36-5) ...
Procesando disparadores para systemd (245.4-4ubuntu3.20) ...
Procesando disparadores para man-db (2.9.1-1) ...
Procesando disparadores para libc-bin (2.31-0ubuntu6) ...
```

2. Apache Service Verification

The Apache service was started with systemctl and its status was reviewed immediately afterward. The service was reported as active (running), confirming that the local HTTP server was operational before requests were generated from Firefox.

```
matthew@matthew-VirtualBox:~$ sudo systemctl start apache2
matthew@matthew-VirtualBox:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-09-15 18:36:00 -05; 2min 32s ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 3611 (apache2)
      Tasks: 55 (limit: 4636)
     Memory: 5.4M
    CGroup: /system.slice/apache2.service
            └─3611 /usr/sbin/apache2 -k start
              └─3613 /usr/sbin/apache2 -k start
                └─3614 /usr/sbin/apache2 -k start

sep 15 18:36:00 matthew-VirtualBox systemd[1]: Starting The Apache HTTP Server...
sep 15 18:36:00 matthew-VirtualBox apachectl[3610]: AH00558: apache2: Could not reliably determine
sep 15 18:36:00 matthew-VirtualBox systemd[1]: Started The Apache HTTP Server.
```

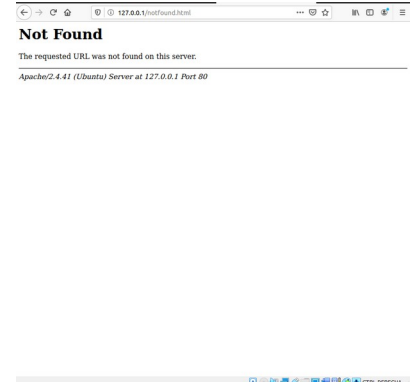
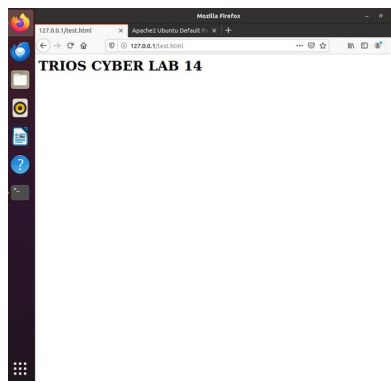
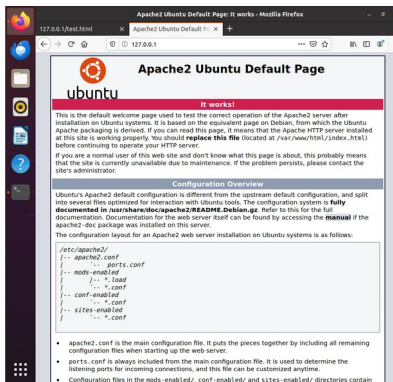
3. Test Page and Access Log Preparation

A local test page was created at /var/www/html/test.html with the heading TRIOS CYBER LAB 14. The web directory was listed to confirm the file, and /var/log/apache2/access.log was truncated so the following log review would focus on traffic generated for this laboratory.

```
matthew@matthew-VirtualBox:~$ echo "<h1>TRIOS CYBER LAB 14</h1>" | sudo tee /var/www/html/test.h
tml
[sudo] contraseña para matthew:
<h1>TRIOS CYBER LAB 14</h1>
matthew@matthew-VirtualBox:~$ ls -l /var/www/html/
total 16
-rw-r--r-- 1 root root 10918 sep 15 18:35 index.html
-rw-r--r-- 1 root root    28 sep 15 18:39 test.html
matthew@matthew-VirtualBox:~$ sudo truncate -s 0 /var/log/apache2/access.log
```

4. Local HTTP Request Generation

HTTP requests were generated from Firefox on the same Ubuntu endpoint using the local address 127.0.0.1. The Apache default page was opened, the custom /test.html page was accessed successfully, and /notfound.html was requested intentionally to generate a Not Found response for later log analysis.



5. Access Log Review and Field Identification

The Apache access log was reviewed with tail. The records exposed the fields required by the exercise: client IP address, HTTP request method, URL/path, status code, and browser user-agent. The local requests originated from 127.0.0.1 and used the GET method. Observed paths included /, /test.html, /notfound.html, /favicon.ico, and /icons/ubuntu-logo.png.

```
matthew@matthew-VirtualBox:~$ sudo tail -n 20 /var/log/apache2/access.log
[sudo] contraseña para matthew:
127.0.0.1 - - [15/Sep/2026:18:51:09 -0500] "GET /test.html HTTP/1.1" 200 311 "-" "Mozilla/5.0 (X
11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:51:09 -0500] "GET /favicon.ico HTTP/1.1" 404 487 "-" "Mozilla/5.0
(X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:51:14 -0500] "GET / HTTP/1.1" 200 3477 "-" "Mozilla/5.0 (X11; Ubu
ntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:51:14 -0500] "GET /icons/ubuntu-logo.png HTTP/1.1" 200 3623 "http:
//127.0.0.1/" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:52:59 -0500] "GET /notfound.html HTTP/1.1" 404 488 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:20 -0500] "GET /notfound.html HTTP/1.1" 404 488 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:20 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:21 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:21 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:21 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:22 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:23 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:25 -0500] "GET / HTTP/1.1" 200 3476 "-" "Mozilla/5.0 (X11; Ubu
ntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:25 -0500] "GET /icons/ubuntu-logo.png HTTP/1.1" 304 180 "http:/
/127.0.0.1/" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:26 -0500] "GET /test.html HTTP/1.1" 304 179 "-" "Mozilla/5.0 (X
11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:27 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:28 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
```

Field	Observed Value	Interpretation
Client IP	127.0.0.1	Local Firefox client
Request method	GET	HTTP requests recorded by Apache
Example path	/test.html	Custom laboratory page
Observed status codes	200, 304, 404	Responses visible in access.log
User-Agent	Mozilla/5.0 ... Firefox/74.0	Firefox on Ubuntu/Linux x86_64

6. GET Request Filtering with grep

The access log was filtered with grep for GET requests. This reduced the review to HTTP request entries and preserved the client address, requested path, response status, and user-agent information for each matching record.

```
matthew@matthew-VirtualBox:~$ sudo grep '"GET ' /var/log/apache2/access.log
127.0.0.1 - - [15/Sep/2026:18:51:09 -0500] "GET /test.html HTTP/1.1" 200 311 "-" "Mozilla/5.0 (X
11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:51:09 -0500] "GET /favicon.ico HTTP/1.1" 404 487 "-" "Mozilla/5.0
(X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:51:14 -0500] "GET / HTTP/1.1" 200 3477 "-" "Mozilla/5.0 (X11; Ubu
ntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:51:14 -0500] "GET /icons/ubuntu-logo.png HTTP/1.1" 200 3623 "http:
//127.0.0.1/" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:52:59 -0500] "GET /notfound.html HTTP/1.1" 404 488 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:20 -0500] "GET /notfound.html HTTP/1.1" 404 488 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:20 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:21 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:21 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:22 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:23 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:25 -0500] "GET / HTTP/1.1" 200 3476 "-" "Mozilla/5.0 (X11; Ubu
ntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:25 -0500] "GET /icons/ubuntu-logo.png HTTP/1.1" 304 180 "http:/
/127.0.0.1/" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:26 -0500] "GET /test.html HTTP/1.1" 304 179 "-" "Mozilla/5.0 (X
11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:27 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:28 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
```

7. Unusual Status Code Identification

A second grep filter selected records containing HTTP status code 404. The output showed requests for /favicon.ico and repeated requests for /notfound.html. The /notfound.html requests corresponded to the intentionally missing resource displayed by Firefox, so the 404 responses were flagged for review as the non-successful status pattern in the sample traffic.

```
matthew@matthew-VirtualBox:~$ sudo grep ' 404 ' /var/log/apache2/access.log
[sudo] contraseña para matthew:
127.0.0.1 - - [15/Sep/2026:18:51:09 -0500] "GET /favicon.ico HTTP/1.1" 404 487 "-" "Mozilla/5.0
(X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:52:59 -0500] "GET /notfound.html HTTP/1.1" 404 488 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:20 -0500] "GET /notfound.html HTTP/1.1" 404 488 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:20 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:21 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:21 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:23 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:27 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:28 -0500] "GET /notfound.html HTTP/1.1" 404 487 "-" "Mozilla/5.
0 (X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:55:42 -0500] "GET /favicon.ico HTTP/1.1" 404 487 "-" "Mozilla/5.0
(X11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
```

8. Repeated Request Detection

Request paths were extracted from the access log with awk, then sorted and counted. The result showed /notfound.html nine times, /icons/ubuntu-logo.png three times, / three times, /test.html twice, and /favicon.ico twice. A focused grep for /test.html confirmed two entries: an initial 200 response and a later 304 response.

```
matthew@matthew-VirtualBox:~$ sudo awk '{print $7}' /var/log/apache2/access.log | sort | uniq -c
| sort -nr
  9 /notfound.html
  3 /icons/ubuntu-logo.png
  3 /
  2 /test.html
  2 /favicon.ico

matthew@matthew-VirtualBox:~$ sudo grep '/test.html' /var/log/apache2/access.log
127.0.0.1 - - [15/Sep/2026:18:51:09 -0500] "GET /test.html HTTP/1.1" 200 311 "-" "Mozilla/5.0 (X
11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
127.0.0.1 - - [15/Sep/2026:18:53:26 -0500] "GET /test.html HTTP/1.1" 304 179 "-" "Mozilla/5.0 (X
11; Ubuntu; Linux x86_64; rv:74.0) Gecko/20100101 Firefox/74.0"
```

9. Web Log Analysis Summary

The filtered evidence was consolidated to summarize the main web-request indicators and the patterns identified during the local test.

Field	Observed Value	Analysis Result
Web service	Apache2 on Ubuntu	Local HTTP test service
Log source	/var/log/apache2/access.log	Apache access records
Client IP	127.0.0.1	Local browser traffic
Request method	GET	Observed request method
Main test path	/test.html	2 requests; 200 then 304
Missing path	/notfound.html	9 repeated requests
Flagged status	404 Not Found	Observed for missing resources
User-Agent	Firefox/74.0 on Ubuntu Linux	Browser recorded in log

10. Conclusions

The local web-log analysis activity was successfully completed on the Ubuntu virtual machine using Apache2, Firefox, and command-line filtering. Requests to the local service produced access-log records containing the client IP, GET method, requested URL/path, HTTP status code, and Firefox user-agent. grep isolated the GET and 404 records, while path counting identified repeated requests, with /notfound.html appearing nine times. The exercise demonstrated how local web logs can be filtered to identify request details, non-successful response codes, and repeated access patterns within an authorized laboratory environment.